

PERSCHOLAS GREATER BOSTON CYBERSECURITY PROGRAM

Module 4: Access Control Lists

Lab Workbook Part 2, Answer Key

*ACL Placement | VTY Lines | Spoofing | ICMP | SNMP | IPv6 ACLs
Network Defense Course | Per Scholas Greater Boston | Cybersecurity Program
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GUIDED LAB Lab 7, Answer Key

ACL Placement, Standard Near Destination, Extended Near Source

Complete Configuration

Task A, Standard ACL: Block HR from Finance server:

```
# Step 1, Create the standard ACL globally first
Router(config)#access-list 10 deny 192.168.10.0 0.0.0.255
Router(config)#access-list 10 permit any

# Step 2, Apply outbound on G0/2, closest to Finance server (destination)
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip access-group 10 out
Router(config-if)#end
Router#write memory
```

Task B, Extended ACL: Block FTP from IT network:

```
# Step 1, Create the extended ACL globally first
Router(config)#ip access-list extended BLOCK-IT-FTP
Router(config-ext-nacl)#deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq 21
Router(config-ext-nacl)#deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq 20
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

# Step 2, Apply inbound on G0/1, closest to IT network (source)
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip access-group BLOCK-IT-FTP in
Router(config-if)#end
Router#write memory
```

Task C, Demonstrate wrong placement, move ACL 10 to G0/0 inbound:

```
# Remove from G0/2 outbound
Router(config)#interface GigabitEthernet0/2
Router(config-if)#no ip access-group 10 out
Router(config-if)#exit
```

```
# Apply inbound on G0/0, WRONG placement
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group 10 in
Router(config-if)#end
# Result: HR is now blocked from reaching IT AND Finance, too broad

# Fix, move it back to correct placement
Router(config)#interface GigabitEthernet0/0
Router(config-if)#no ip access-group 10 in
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip access-group 10 out
Router(config-if)#end
```

Verification Output

```
Router#show access-lists
Standard IP access list 10
 10 deny 192.168.10.0 0.0.0.255
 20 permit any

Extended IP access list BLOCK-IT-FTP
 10 deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq ftp
 20 deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq ftp-data
 30 permit ip any any

Router#show ip interface GigabitEthernet0/2
Outbound access list is 10

Router#show ip interface GigabitEthernet0/1
Inbound access list is BLOCK-IT-FTP
```

Expected Outcome

Test	Expected Result	Answer
HR-PC pings Finance Server	Fails, standard ACL blocks HR outbound on G0/2	✓ PASS
HR-PC pings IT-PC	Succeeds, standard ACL on G0/2 does not affect this path	✓ PASS
IT-PC accesses Finance Server on port 21	Fails, extended ACL blocks FTP inbound on G0/1	✓ PASS
IT-PC pings Finance Server	Succeeds, only FTP is blocked, ICMP is permitted	✓ PASS
Task C, HR-PC pings IT-PC with ACL on G0/0	Fails, wrong placement blocks more than intended	✓ PASS

Reflection Answers

Question 1: Why did moving the standard ACL to G0/0 inbound block HR from reaching IT?

Answer: Standard ACLs only filter based on source IP address, they cannot distinguish between destinations. When the ACL was placed on G0/0 inbound, it checked every packet entering from the HR network and denied it regardless of where it was going, Finance or IT made no difference. By moving it to G0/2 outbound, the ACL only activates when traffic is about to exit toward the Finance network specifically, leaving all other paths unaffected.

Question 2: Why is the extended ACL placed inbound on G0/1 instead of outbound on G0/2?

Answer: Extended ACLs should always be placed as close to the source as possible to stop unwanted traffic before it wastes bandwidth crossing the network. G0/1 is the interface directly connected to the IT network, the source of the FTP traffic. Placing it here means FTP packets are dropped immediately at entry. If it were placed on G0/2 outbound instead, the FTP traffic would still travel all the way from IT across the network to G0/2 before being dropped, wasting bandwidth unnecessarily.

GUIDED LAB Lab 8, Answer Key

Apply ACL to Interface and VTY Lines

Complete Configuration

Step 1, Enable SSH on the router:

```
Router(config)#hostname R1
R1(config)#ip domain-name perscholas.org
R1(config)#crypto key generate rsa modulus 1024
R1(config)#username admin secret cisco123
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
```

Step 2, Create and apply standard ACL to VTY lines:

```
R1(config)#ip access-list standard SSH-ONLY
R1(config-std-nacl)#permit host 192.168.1.10
R1(config-std-nacl)#deny any
R1(config-std-nacl)#exit

R1(config)#line vty 0 4
R1(config-line)#access-class SSH-ONLY in
R1(config-line)#exit
```

Step 3, Create and apply named extended ACL on internet-facing interface:

```
R1(config)#ip access-list extended INTERNET-IN
R1(config-ext-nacl)#permit tcp any 192.168.1.0 0.0.0.255 eq 80
R1(config-ext-nacl)#permit tcp any 192.168.1.0 0.0.0.255 eq 443
R1(config-ext-nacl)#deny ip any any
R1(config-ext-nacl)#exit
```

```

R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip access-group INTERNET-IN in
R1(config-if)#end
R1#write memory

```

Step 4, Remove ACL from interface without deleting it:

```

# Remove from interface only, ACL still exists on router
R1(config)#interface GigabitEthernet0/1
R1(config-if)#no ip access-group INTERNET-IN in
R1(config-if)#exit

# Verify ACL still exists on router
R1#show access-lists
Extended IP access list INTERNET-IN
 10 permit tcp any 192.168.1.0 0.0.0.255 eq 80
 20 permit tcp any 192.168.1.0 0.0.0.255 eq 443
 30 deny ip any any

# Now delete it completely
R1(config)#no ip access-list extended INTERNET-IN
R1#show access-lists
# No output, ACL is completely gone

```

Expected Outcome

Test	Expected Result	Answer
Mgmt-PC SSH to Router	Succeeds, explicitly permitted by SSH-ONLY ACL	✓ PASS
PC1 SSH to Router	Fails, blocked by deny any in SSH-ONLY ACL	✓ PASS
Internet Host sends HTTP to inside network	Succeeds, port 80 permitted by INTERNET-IN	✓ PASS
Internet Host sends Telnet to inside network	Fails, not permitted by INTERNET-IN	✓ PASS
After no ip access-group, show access-lists	ACL still exists, only unapplied from interface	✓ PASS
After no ip access-list, show access-lists	ACL completely gone from router	✓ PASS

Reflection Answer

Question: What is the difference between ip access-group and ip access-class?

Answer: ip access-group is used to apply an ACL to a physical interface, it filters traffic entering or leaving that interface. ip access-class is used specifically for VTY lines, it controls who is allowed to remotely access the router itself using SSH or Telnet. The key difference is the target: ip access-group protects traffic passing through the router, while ip access-class protects administrative access to the router itself.

Mitigate IP Spoofing Attacks

Complete Configuration

Step 1, Anti-spoofing ACL on internet-facing interface:

```
Router(config)#ip access-list extended ANTI-SPOOF-INTERNET

# Block all-zeros addresses
Router(config-ext-nacl)#deny ip 0.0.0.0 0.255.255.255 any

# Block loopback addresses
Router(config-ext-nacl)#deny ip 127.0.0.0 0.255.255.255 any

# Block APIPA addresses
Router(config-ext-nacl)#deny ip 169.254.0.0 0.0.255.255 any

# Block RFC 1918 private ranges
Router(config-ext-nacl)#deny ip 10.0.0.0 0.255.255.255 any
Router(config-ext-nacl)#deny ip 172.16.0.0 0.15.255.255 any
Router(config-ext-nacl)#deny ip 192.168.0.0 0.0.255.255 any

# Block multicast arriving as unicast
Router(config-ext-nacl)#deny ip 224.0.0.0 15.255.255.255 any

# Permit all legitimate traffic
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

Router(config)#interface Serial0/0/0
Router(config-if)#ip access-group ANTI-SPOOF-INTERNET in
Router(config-if)#end
```

Step 2, ACL to block spoofed internal addresses arriving from outside:

```
Router(config)#ip access-list extended ANTI-SPOOF-INTERNAL

# Only permit packets genuinely from the internal network
Router(config-ext-nacl)#permit ip 192.168.1.0 0.0.0.255 any

# Block everything else
Router(config-ext-nacl)#deny ip any any
Router(config-ext-nacl)#exit

Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group ANTI-SPOOF-INTERNAL in
Router(config-if)#end
Router#write memory
```

Expected Outcome

Test	Expected Result	Answer
Legitimate internet traffic reaches internal network	Succeeds, permit ip any any allows it through	✓ PASS
Packet with source 192.168.x.x arrives from internet	Fails, RFC 1918 address blocked	✓ PASS
Packet with source 127.x.x.x arrives from internet	Fails, loopback address blocked immediately	✓ PASS
Packet with source 10.x.x.x arrives from internet	Fails, RFC 1918 private range blocked	✓ PASS
Internal PC sends traffic outbound	Succeeds, 192.168.1.0/24 permitted by ANTI-SPOOF-INTERNAL	✓ PASS

Reflection Answer

Question: Why would a packet from 192.168.1.50 arriving on the internet-facing interface be considered spoofed?

Answer: The address 192.168.1.50 is a private RFC 1918 address, these addresses are only used inside private networks and are never routed across the public internet. If a packet claiming to come from 192.168.1.50 arrives on the internet-facing interface, it means someone has faked that source address, because that address could never legitimately travel across the internet to reach the router. This is a clear indicator of IP spoofing and must be dropped immediately.

PRACTICE LAB Lab 10, Answer Key

Permit Necessary Services Through a Firewall

Complete Configuration, FIREWALL-POLICY ACL

```
Router(config)#ip access-list extended FIREWALL-POLICY

# Policy 1, Permit DNS to DNS server
Router(config-ext-nacl)#permit udp any host 192.168.1.50 eq 53

# Policy 2, Permit SMTP to mail server
Router(config-ext-nacl)#permit tcp any host 192.168.1.60 eq 25

# Policy 3, Permit FTP to mail server
Router(config-ext-nacl)#permit tcp any host 192.168.1.60 eq 21
Router(config-ext-nacl)#permit tcp any host 192.168.1.60 eq 20

# Policy 4, Permit SSH only from management station
Router(config-ext-nacl)#permit tcp host 10.0.0.5 any eq 22

# Policy 5, Permit Syslog from router interface only
Router(config-ext-nacl)#permit udp host 209.165.200.1 any eq 514

# Policy 6, Permit SNMP traps from management station only
```

```

Router(config-ext-nacl)#permit udp host 10.0.0.5 any eq 162

# Implicit deny blocks everything else
Router(config-ext-nacl)#exit

# Apply inbound on internet-facing interface
Router(config)#interface Serial0/0/0
Router(config-if)#ip access-group FIREWALL-POLICY in
Router(config-if)#end
Router#write memory

```

Verification Output

```

Router#show access-lists FIREWALL-POLICY
Extended IP access list FIREWALL-POLICY
 10 permit udp any host 192.168.1.50 eq domain
 20 permit tcp any host 192.168.1.60 eq smtp
 30 permit tcp any host 192.168.1.60 eq ftp
 40 permit tcp any host 192.168.1.60 eq ftp-data
 50 permit tcp host 10.0.0.5 any eq 22
 60 permit udp host 209.165.200.1 any eq syslog
 70 permit udp host 10.0.0.5 any eq snmptrap

Router#show ip interface Serial0/0/0
Inbound access list is FIREWALL-POLICY

```

Expected Outcome

Test	Expected Result	Answer
DNS traffic to 192.168.1.50 port 53	Succeeds, DNS explicitly permitted	✓ PASS
SMTP traffic to 192.168.1.60 port 25	Succeeds, SMTP explicitly permitted	✓ PASS
SSH from 10.0.0.5 to router	Succeeds, management station permitted for SSH	✓ PASS
SSH from any other host to router	Fails, only management station is permitted	✓ PASS
HTTP traffic from internet to inside	Fails, HTTP not in the permitted services list	✓ PASS
Telnet from internet to any inside host	Fails, Telnet not in the permitted services list	✓ PASS

Reflection Answer

Question: Why is it safer to explicitly permit only needed services rather than explicitly deny unwanted ones?

Answer: The explicit deny approach requires you to know every possible attack and every possible bad service in advance, and block them all individually. If attackers find a new attack vector you did not think to block, it gets through. The explicit permit approach works the opposite way, you define exactly what is allowed and deny everything else automatically through implicit deny. Even if a completely new attack type appears, it will be

blocked because it was never explicitly permitted. This is the safer strategy because it limits access by default rather than trying to predict every threat.

PRACTICE LAB Lab 11, Answer Key

Mitigate ICMP and SNMP Attacks

Complete Configuration

Task A, ICMP ACL on Serial0/0/0 inbound (internet-facing):

```
Router(config)#ip access-list extended ICMP-INTERNET-IN

# Permit echo-reply, lets inside users receive ping responses
Router(config-ext-nacl)#permit icmp any any echo-reply

# Permit source-quench, congestion control messages
Router(config-ext-nacl)#permit icmp any any source-quench

# Permit unreachable, informs senders when ACL blocks their packets
Router(config-ext-nacl)#permit icmp any any unreachable

# Block all other ICMP types from entering
Router(config-ext-nacl)#deny icmp any any

# Permit all non-ICMP traffic
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

Router(config)#interface Serial0/0/0
Router(config-if)#ip access-group ICMP-INTERNET-IN in
Router(config-if)#exit
```

Task A, ICMP ACL on GigabitEthernet0/0 inbound (internal-facing):

```
Router(config)#ip access-list extended ICMP-INTERNAL-IN

# Permit echo, allows inside users to ping external hosts
Router(config-ext-nacl)#permit icmp any any echo

# Permit parameter-problem, packet header error messages
Router(config-ext-nacl)#permit icmp any any parameter-problem

# Permit packet-too-big, MTU discovery
Router(config-ext-nacl)#permit icmp any any packet-too-big

# Permit source-quench, congestion throttle
Router(config-ext-nacl)#permit icmp any any source-quench

# Block all other ICMP types leaving the internal network
Router(config-ext-nacl)#deny icmp any any
```



```
# Permit all non-ICMP traffic
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group ICMP-INTERNAL-IN in
Router(config-if)#exit
```

Task B, Disable SNMP completely:

```
Router(config)#no snmp-server
Router#show snmp
# SNMP is not enabled, no output
```

Task C, Restrict SNMP to management station only:

```
Router(config)#ip access-list standard SNMP-ACCESS
Router(config-std-nacl)#permit host 192.168.1.50
Router(config-std-nacl)#deny any
Router(config-std-nacl)#exit

Router(config)#snmp-server community public ro SNMP-ACCESS
Router#write memory
```

Expected Outcome

Test	Expected Result	Answer
Inside PC pings Internet Host	Succeeds, echo permitted outbound from inside	✓ PASS
Internet Host pings Inside PC	Fails, echo blocked inbound on Serial0/0/0	✓ PASS
Inside PC receives ping reply from internet	Succeeds, echo-reply permitted inbound	✓ PASS
SNMP query from Mgmt Station	Succeeds, management station is permitted	✓ PASS
SNMP query from any other host	Fails, blocked by SNMP-ACCESS ACL	✓ PASS
show snmp after no snmp-server	SNMP disabled, no response to any query	✓ PASS

Reflection Answer

Question: Why is it unsafe to completely block all ICMP traffic?

Answer: Completely blocking all ICMP would break several essential network functions. Without echo-reply, inside users could not receive ping responses and would be unable to test connectivity. Without unreachable messages, senders would never know when their packets are being blocked and troubleshooting would become nearly impossible. Without packet-too-big messages, MTU discovery breaks and packets get fragmented or dropped silently. Without source-quench, there is no congestion signaling mechanism. The correct approach is to permit only the specific ICMP types the network needs and block the rest, not block everything.

IPv6 ACL Configuration, Challenge Lab

Complete Configuration, LAN-ONLY IPv6 ACL

```
# Step 1, Create the named IPv6 ACL
Router(config)#ipv6 access-list LAN-ONLY

# Permit traffic from the internal IPv6 network
Router(config-ipv6-acl)#permit ipv6 2001:DB8:ACAD:1::/64 any

# Explicitly permit NDP Neighbor Advertisement messages
Router(config-ipv6-acl)#permit icmp any any nd-na

# Explicitly permit NDP Neighbor Solicitation messages
Router(config-ipv6-acl)#permit icmp any any nd-ns

# Deny all other IPv6 traffic
Router(config-ipv6-acl)#deny ipv6 any any
Router(config-ipv6-acl)#exit

# Step 2, Apply inbound on G0/0 using ipv6 traffic-filter
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ipv6 traffic-filter LAN-ONLY in
Router(config-if)#end
Router#write memory
```

Verification Output

```
Router#show ipv6 access-lists
IPv6 access list LAN-ONLY
  permit ipv6 2001:DB8:ACAD:1::/64 any sequence 10
  permit icmp any any nd-na sequence 20
  permit icmp any any nd-ns sequence 30
  deny ipv6 any any sequence 40

Router#show ipv6 interface GigabitEthernet0/0
  Inbound access list LAN-ONLY

Router#show running-config | section ipv6 access-list
ipv6 access-list LAN-ONLY
  permit ipv6 2001:DB8:ACAD:1::/64 any
  permit icmp any any nd-na
  permit icmp any any nd-ns
  deny ipv6 any any
```

Expected Outcome

Test	Expected Result	Answer
Inside PC sends IPv6 traffic outbound	Succeeds, 2001:DB8:ACAD:1::/64 is permitted	✓ PASS
Outside Host sends IPv6 traffic inbound	Fails, not in the permitted prefix, implicit deny	✓ PASS
NDP works between devices on same segment	Succeeds, nd-na and nd-ns explicitly permitted	✓ PASS
show ipv6 access-lists shows LAN-ONLY	ACL visible with all four ACEs and sequence numbers	✓ PASS
show ipv6 interface G0/0	LAN-ONLY listed as inbound IPv6 access list	✓ PASS

Reflection Answers

Question 1: What happens if you configure deny ipv6 any any without first permitting NDP?

Answer: IPv6 communication on the network stops working completely. NDP (Neighbor Discovery Protocol) uses ICMPv6 nd-na and nd-ns messages for devices to discover each other's MAC addresses, it is the IPv6 equivalent of ARP in IPv4. Without NDP working, devices cannot resolve IPv6 addresses to MAC addresses, which means they cannot send any IPv6 traffic even if the ACL permits the traffic itself. The deny ipv6 any any at the bottom blocks the NDP messages along with everything else unless you explicitly permit them first.

Question 2: Why does IPv6 use ipv6 traffic-filter instead of ip access-group?

Answer: Cisco IOS uses separate commands for IPv4 and IPv6 to clearly distinguish between the two protocol stacks. ip access-group is specifically for IPv4 ACLs, it only processes IPv4 traffic. ipv6 traffic-filter is specifically for IPv6 ACLs, it only processes IPv6 traffic. Both commands can be applied to the same interface at the same time to filter both protocol stacks simultaneously, which is essential in dual-stack environments.

Question 3: What is the key difference between ip access-list and ipv6 access-list?

Answer: IPv4 ACLs can be either numbered or named, you can create access-list 10 or ip access-list standard MYACL. IPv6 ACLs must always be named, there are no numbered IPv6 ACLs at all. This means every IPv6 ACL must have a descriptive name, which actually encourages better documentation practices compared to IPv4 where numbered ACLs are common.

End of Module 4 Lab Answer Key, Part 2

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